Minimap: An Interactive Dynamic Decision Making Game for Search and Rescue Missions

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ARTICLE INFO

Keywords:
dynamic decision making
microworlds
human experiments
search and rescue missions

ABSTRACT

Many aspects of human dynamic decision making (DDM) behaviors can be studied with computer-simulated games, called microworlds. However, most microworlds only emphasize some aspects of DDM and are not flexible in terms of generating different decision situations. In this paper, we introduce *Minimap*, a dynamic interactive game for studying DDM in search and rescue missions, which incorporates all the essential characteristics of DDM, and offers flexibility regarding experimental setups and manipulating experiment scenarios. We present an experiment constructed with Minimap to demonstrate the development of scenarios that vary in structural complexity (simple vs. complex) and opaqueness (field of view-FoV vs. map view) under time constraints. We found that: humans performed more effectively and efficiently with the Map view than with FoV; humans made fewer redundant movements in the Map view compared to FoV; human learning in terms of effectiveness and efficiency was not affected by structural complexity, but it impacted their redundancy over time; and human's coverage was not affected even when opaqueness and structural complexity increase. These findings demonstrate how Minimap can be used to investigate many aspects of DDM behavior and inform the design of scenarios that have implications for building advisory support systems in search and rescue missions.

1. Introduction

The real world is full of dynamic decision-making (DDM) tasks, which are essentially a process that involves a series of sequential and interdependent decisions made in an environment that changes independently and also as a consequence of those decisions (Brehmer, 1992). A task is designated as a DDM task if it possesses four characteristics, namely *dynamics* (the dependence of the system's state on its state at an earlier time), *complexity* (the number of components in the system, the number of relationships among them and the nature of these relationships), *opaqueness* (the invisibility of some aspects of the system) and *dynamic complexity* (the inclusion of feedback delays and interrelationships of decisions over time) (Gonzalez, 2017). Examples of DDM tasks include rescuers in an unknown environment who must determine which action to take while navigating and searching for victims, drivers trying to find the best route in heavy traffic, investors aiming a determining the best portfolio in a changing stock market, or individuals trying to allocate limited resources under time constraints and workload.

Making decisions in such DDM environments presents a great challenge to humans. Indeed, previous findings suggest that humans cannot always improve their decisions even with practice in a task (Brehmer, 1980), and that they generally have difficulty processing feedback delays (Brehmer, 1992) and associating their actions with the results of their decisions over time (Gonzalez, Lerch and Lebiere, 2003). People's performance is particularly deteriorating in situations involving time constraints (Gonzalez, 2004; Kerstholt and Raaijmakers, 2002) and workload (Gonzalez, Thomas and Vanyukov, 2005a). In addition, humans struggle to process and remember rare events, which occur infrequently (Hertwig, Barron, Weber and Erev, 2004). Past research also indicate that people suffer from a number of biases that depend on their own experience with a sequence of decisions; for example, they make choices based

Part of the results reported here have been presented at the AAAI Fall Symposium 2021: Computational Theory of Mind for Human-Machine Teams (Gulati, Nguyen and Gonzalez, 2021).

Transparency and Openness: Minimap is provided as a free and open-source application, which can be accessed at `https://github.com/DDM-Lab/MinimapInteractiveDDMGame`. All human data and all scripts used for the analyses presented in this manuscript are available on Github: `https://github.com/DDM-Lab/HumanExperimentMinimap`.

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on recent experiences (Gonzalez and Dutt, 2011b), and have tendency to observe patterns in data where there are none (West and Lebiere, 2001; Gilovich, Vallone and Tversky, 1985).

Over the years, researchers have addressed some of these challenges and have improved our understanding of the process by which humans learn to make decisions in those situations (Gonzalez, 2017). Since the study of human decision behavior in naturalistic settings is complex, a large body of DDM research has addressed questions with computer-interactive simulations, often referred to as "microworlds" (Brehmer and Dörner, 1993; Gonzalez, Vanyukov and Martin, 2005b). One of the key benefits of using microworlds in DDM research is that it allows researchers to conduct experimental human research within complex dynamic decision-making situations while maintaining the essential features of dynamic tasks and experimental control (Funke, 1995).

To date, a large variety of microworlds have been developed in an effort to study human performance in numerous complex control task settings, including military command and control (Marusich, Bakdash, Onal, Yu, Schaffer, O'Donovan, Höllerer, Buchler and Gonzalez, 2016), resource allocation (Gonzalez, 2005), supply chain management (Sterman, 1989; Cronin, Gonzalez and Sterman, 2009), and dynamic stock and flow systems (Gonzalez and Dutt, 2011a). Unfortunately, most microworlds represent and allow researchers to study only some aspects of DDM tasks, such as uncertainty or complexity, but they do not support the study of other essential features, such as time constraints, opaqueness, real-time decision processes, interdependence between decisions, or their effects over time (dynamic complexity) (Gonzalez et al., 2005b). Quite conversely, many of the tasks in search and choice direction are constructed in a structurally simple fashion (i.e. few alternatives, where time constraints, dynamic complexity, and other types of uncertainty in dynamic tasks are not represented (Lejarraga, Hertwig and Gonzalez, 2012; Mehlhorn, Ben-Asher, Dutt and Gonzalez, 2014). Notwithstanding the increasing use of microworlds in DDM research, it is clear that little research has been dedicated to addressing the flexibility and generality of microworlds. A large portion of microworlds are tied to specific scenarios with no flexibility to generate various decision situations. As suggested by previous research (Gonzalez and Dutt, 2011a), there is significant room left for the development and investigation of more generic and flexible DDM microworlds that can be used for research.

The characteristics of dynamic systems are present in a large number of naturalistic decision-making tasks (Funke, 1995; Klein, 2017). One particular example, which we consider here, is the domain of urban search and rescue. In disaster environments, one inevitably must cope with the multiple possibilities and alternatives and weigh multiple attributes (complexity) accompanied by a great deal of uncertainty about the situation (opaqueness). Furthermore, long feedback delays, which are often common, generate dynamic complexity, as it is difficult to associate actions with outcomes over time (Gonzalez, Fakhari and Busemeyer, 2017). In addition, time constraints and the interrelationships of decisions over time are inherent in the process of search and rescue of victims in a real-time evolution of the environment. Since these features are illustrative of DDM environments (Omodei and Wearing, 1995), numerous studies on robotics, artificial intelligence (AI), and human machine teaming have been contextualized in the search and rescue domain, with the use of microworlds. For instance, FireChief is a computer-simulated microworld in which a decision maker as chief fire officer has to control the spread of fires by appropriately allocating firefighting appliances at their disposal (Omodei and Wearing, 1995). USARSim is another search and rescue simulation game in which the team explores an unknown environment to identify as many positions of victims as possible (Lewis, Wang, Chien, Velagapudi, Scerri and Sycara, 2011). In the same vein, Corral, Tatapudi, Buchanan, Huang and Cooke (2021); Lematta, Coleman, Bhatti, Chiou, McNeese, Demir and Cooke (2019); Lematta et al. (2019) used a search and rescue task environment simulated in the Minecraft video game to study whether an artificial system could generate the theory of mind of human teammates.

Despite such advances in the development of computer-based microworlds and video game simulations for behavioral science and AI, the potential of browser-based microworlds and on-line data collection has been limited. In fact, the sophisticated technical restrictions of video games hamper the exchange, reuse, customization, and extension of frameworks, whereas the desktop-based applications limit the data collection and reduce flexibility in conducting human experiments at different time intervals. With that, there is a need to leverage browser-based experiments, given that they are suitable for real-time gaming, offer great flexibility with regard to experimental setups, and an ubiquitous communication tool for data collection that can take place in a multitude of settings (Woods, Velasco, Levitan, Wan and Spence, 2015; Mathôt, Schreij and Theeuwes, 2012).

To this end, our goal here is to tap into this potential and to increase the accessibility and usefulness of a browser-based microworld for online behavioral research by introducing *Minimap*, a dynamic interactive game for studying decision-making behavior in search and rescue missions. Minimap has all the essential characteristics of a DDM task, since it presents a decision maker with a sequential decision problem in which the environment evolves

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both autonomously and in response to their decision (*dynamics*). The decision maker has to make decisions within time constraints in a complex topological structure and multiple rescue mechanism settings (*complexity*) coupled with uncertainty about the environment (*opaqueness*) and long feedback delays regarding the immediate exploratory decisions (*dynamic complexity*). The simulated environment in Minimap has by far a larger dimension compared to any typical 2D gridworlds in AI research (Sutton and Barto, 2018). More importantly, unlike existing simulated search and rescue tasks, Minimap is designed to make use of the full capabilities of modern browsers for interactivity: it offers flexibility to experimenters in the definition of experimental setups, manipulating experiment scenarios, varying different aspects of DDM situations, and collecting human data. In general, Minimap is built to be fully customizable and extensible by advanced users.

In this paper, we present Minimap and describe all its potential to study the characteristics of DDM in search and rescue missions. We also demonstrate the use of Minimap in an online experiment, where we investigate the effect of structural complexity in association with opaqueness (i.e. uncertainty and ambiguity about the environment) on human decision behavior. The results indicate that Minimap can be used to investigate many aspects of DDM behavior, and our findings can inform the design of scenarios that may have implications for training and building an advisory support system in search and rescue missions.

2. Minimap: A Game to Study Search and Rescue Missions

Minimap is implemented to take into account all the characteristics of a DDM task with the purpose of studying human learning and decision making in a generic search and choice task. Its design is aimed to be fully customizable and extensible by advanced users. More importantly, Minimap is built to enable experimenters to systematically manipulate or vary different aspects of complex and dynamic situations.

In the following, we first present a generic version of the search and rescue task that is designed to simulate typical task constraints that a rescuer might encounter[1]. Then we describe Minimap's architecture and components in a study that can be customized through Minimap configuration to fit the requirements of the scientific endeavor at hand. Finally, we explain how Minimap features all four essential characteristics of DDM environments (Gonzalez et al., 2005b; Gonzalez, 2017).

2.1. A Search and Rescue Scenario: Generic Instructions

The goal of the mission is to obtain as many points as possible by finding and rescuing victims from inside a building that collapses within a 5-minute time limit. The layout of the game is illustrated in Fig. 1. That is, the rescuer must navigate through an environment and avoid obstacles to locate and save victims. Throughout the mission, the rescuer is informed about the time left in an episode (i.e., five-minute mission) as well as the accumulated points (i.e., "Points") by rescuing the victims during each of the episodes.

The environment of the Minimap task is a 93 x 50 grid of cells, which represents one floor of this building. In each square (cell), the rescuer may find nothing, an obstacle, or a victim. There are 34 victims scattered throughout the building, of which 10 are severely injured and need more urgent assessment, while the other 24 victims are considered not critically injured. This feature is simulated as critical victims and regular victims (high and low rewards). Concretely, regular victims are worth ten points each, are represented by green squares, they take shorter time (five button presses) to save, and are available to be rescued throughout the entire mission. By contrast, critical victims are worth thirty points each, are represented by yellow squares, require a longer time (ten button presses) to be rescued, and they disappear from the building one minute before the mission is over (i.e., they die).

Additionally, there are many obstacles and walls placed in the path of the victims, forcing the rescuer to look for alternative routes. Moreover, rooms have closed doors, which require the rescuer to open to enter the room. To save victims or open doors, the rescuer must be in a square right next to the victim or the door. See Figure A2 in Appendix for detailed keyboard controls of the game.

The rescuer is always able to see objects within a 5x5 square area around them, "vision area", and they may be able to additionally observe the walls in the building depending on the degree of environmental visibility. Structural complexity is dictated by the number of obstacles and the difficulty to reach the critical victims of the building. Therefore, the number of obstacles and walls and the arrangement of critical victims vary depending on the levels of complexity (see Section 3.1.1 for more details).

[1] Demo of the game: `http://halle.psy.cmu.edu:5701/demo/`

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Figure 1: The graphical user interface of Minimap for a search and rescue task with the full view mode. The red dot represents the current position of human player in the game. Green and yellow squares represent unsaved victims, grey squares represent walls, light grey squares represent obstacles, and purple squares are designated for doors. The wall and obstacle are not passable while the empty blue squares are areas in which the player can traverse.

2.2. Characteristics of a DDM Task in Minimap

2.2.1. Dynamics

Minimap embodies temporal dependencies among system states, and the environment is dynamic with respect to participants' activity, since one does change the situation itself with subsequent activity taken in the environment (at time period t) is partly dictated by previous activity (time period $t-1$). This is one of the fundamental characteristics of dynamic systems (Edwards, 1962). In a search and rescue mission with Minimap illustrated in Fig. 1, the environment evolves both autonomously (e.g., victims die and disappear from the environment in the absence of any decision input from human participants) and in response to the individual's decision (e.g., the environment changes once a door is opened, or a victim is saved and no longer injured).

The degree of dynamics can be easily varied in Minimap by making the environment change continually or at certain intervals; for example, we can manipulate the time victims die to make the disappearance (death) of victims occur every two minutes.

2.2.2. Complexity

As pointed out in previous research (Schmid, Ragni, Gonzalez and Funke, 2011; Gonzalez et al., 2017), complexity is a relative term and can arise from various sources. In cognitive and AI research, complexity has mostly been studied as an issue wherein a task possesses a large number of interrelated elements to be processed within a unit of time (Osman, 2010; Madhavan, Gonzalez and Lacson, 2007). Complexity can also arise from the tension between the costs and benefits of the decision (Nguyen and Gonzalez, 2020), or from the presence of multiple alternative decisions with multiple attributes (Schmid et al., 2011).

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Minimap offers great flexibility in manipulating this kind of complexity. In particular, a different number of components can be added to the environment, as well as the placement of them, which is called *structural* or *environmental complexity*. In addition, time constraints or the time window in which victims vanish are also variables in Minimap.

Moreover, a scheme of rewards and costs is readily configurable in Minimap. Take the search and rescue scenario in Fig. 1 as an example; the yellow and green squares represent critical victims and regular victims who give high and low rewards/costs, respectively. Critical victims are designed to be worth more points than regular victims, yet they require a longer time (more effort) to be rescued. With that, one can presumably create any kind of mission and scenario that not only controls structural complexity but also decision complexity (i.e., conflict between benefit and cost in gridworld task (Nguyen and Gonzalez, 2020)).

2.2.3. Opaqueness

Evidently, a task becomes more challenging in the presence of uncertainty about the environment. Therefore, the next essential element of DDM tasks is opaqueness that arises when there are some aspects of the environment is invisible to a decision maker. Often, it is represented by the availability of situational information that individuals gain through their learning about the state of the world Todd, Hills and Robbins (2012); Mehlhorn, Newell, Todd, Lee, Morgan, Braithwaite, Hausmann, Fiedler and Gonzalez (2015).

This opaqueness characteristic is fully incorporated and customizable in Minimap at different levels of granularity through the visibility variable in its configuration. More concretely, the degree of opaqueness can vary in a way that the user can observe part of the environment within their restricted vision area (field of view - FoV); or they can view the structure of the map (see Fig. 3 and Section 3.1.1 for more detail). The opaqueness is also featured in Minimap by the fact that the information on the number of components in the Minimap environment is completely concealed from the user. As a result, they must explore by interacting with the environment and gradually learn about these unknown parts of the environment.

2.2.4. Dynamic Complexity

Dynamic complexity varies in their inclusion of delayed feedback (Diehl and Sterman, 1995). Previous research has shown that increased feedback delays between decision actions and the corresponding decision outcomes negatively affect human performance in a variety of dynamic tasks (Brehmer, 1995; Gonzalez, 2017).

Feedback delays are inherent in Minimap, a sequential decision-making task. That is, the user has to make a sequence of decisions without feedback, knowing the outcome only at the end of the sequence. For example, in the search and rescue scenario, a decision maker must navigate the space making decisions on which actions to take while being unaware of where victims and obstacles are located. Also, the large dimension of the environment space makes for long feedback delays. The final result is also delayed until the end of the game duration.

Similar to the other characteristics, the feature of feedback delays can be readily altered in Minimap. For example, the delay in the individual's action can be controlled by the variable `delayed time` defined by the configuration time. Essentially, the variable adds a constant number of time periods k and the action only takes effect after the time periods k have passed.

2.3. Technical Underpinnings

In this section, we introduce the Minimap technical architecture to give a basic idea of how it has been developed, and how to customize different DDM elements of an experiment.

2.3.1. Minimap's Architecture

As illustrated in Fig. 2, Minimap is designed relying on a client-server architecture consisting of two parts: client browsers and a web server. On the Minimap client side, it is based on open Web technologies (HTML5, CSS, JavaScript) as well as integrating third-party libraries such as jQuery[2] and p5.js[3] for graphical design. It is also equipped with Socket-io client library[4] for facilitating real-time interaction with a web browser or other client applications. On the server side, Minimap is built upon FastAPI[5] web framework in Python. Likewise, the Socket.io library for the server is also incorporated to enable the interactions with browser clients. For communication and exchange of data, Minimap's

[2]a cross-browser JavaScript library designed to simplify the client-side manipulation of an HTML page (`https://jquery.com/`).
[3]a JavaScript library for drawing functionality using the HTML5 canvas element (`https://p5js.org/`).
[4]technology that facilitates two-way interactive communication session between the user's browser and a server (`https://socket.io/`).
[5]a modern, high-performance, web framework for building APIs with Python 3.6+ (`https://fastapi.tiangolo.com/`).

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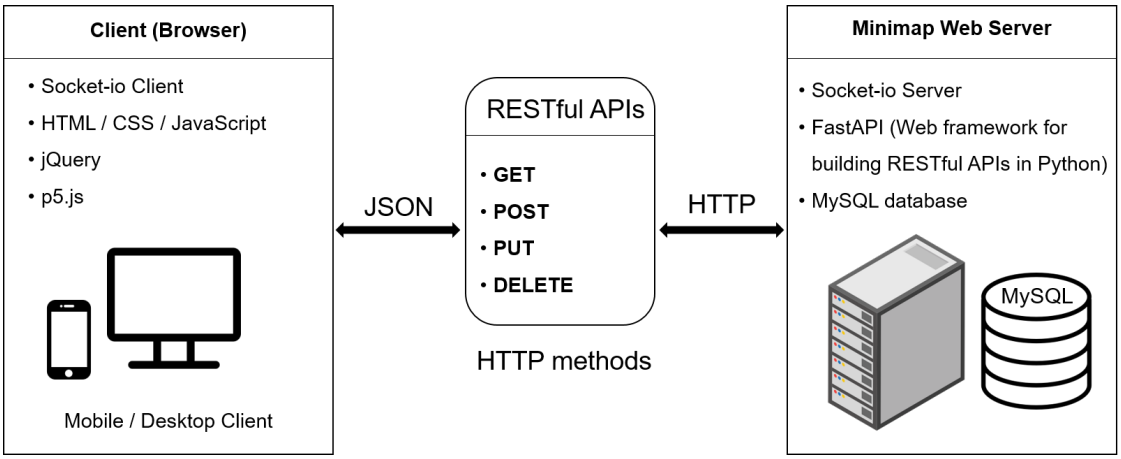

Figure 2: Schematic representation of the Minimap architecture. The two main components: Web server and client. Client sends a request in form of JSON data and Server sends a response through RESTful APIs methods.

architecture adopts RESTful API web service to take advantage of REST APIs' benefits, which are well known for scalability and flexibility (i.e., an application using REST APIs can be scaled quickly primarily due to the separation between the client and the server). Finally, Minimap uses the MySQL database management system to store data collected from human subjects. Trajectories of humans playing during the game are recorded once there is an event occurrence executed by human participants. Otherwise, human trajectories are automatically saved every 30 seconds regardless of any input from them.

Overall, Minimap is designed to be cross-browser and platform agnostic. It is able to facilitate online participation because users can join an experiment directly in the browser window without the need of installing any additional software. This feature distinguishes Minimap from other search and rescue tasks simulated in the Minecraft video game (Corral et al., 2021; Lematta et al., 2019; Freeman, Huang, Wood and Cauffman, 2021).

2.3.2. Customizing Experiments with Minimap

Minimap provides flexibility to adjust experiment variables to study different aspects of DDM tasks. In particular, the experiment settings are specified in the file `mission/config.json`. This file contains the variables needed to define the functioning of the experiment as follows:

```
{
    "load_new_map": 1,
    "map_file": "map_design.csv",
    "map_dir": "mission/static/data/",
    "game_duration": 300,
    "max_episode": 8,
    "victim_die": 120,
    "visibility": "full",
    "complexity": "complex",
    "delayed_time": 1000
}
```

Interface Modification. Minimap, by default, features the environment as illustrated in Fig. 1 provided that the environment design is customizable. That is, one can simply customize the Minimap layouts including the location, and the number of objects presented in the map (e.g., victims, walls, obstacles, and doors) to create different environment. Technically, it can be achieved without the need for specialized technical knowledge: by modifying the base map designed in the Excel file, which is placed in the folder (`mission/static/map_design.xlsx`) and then saving it as a CSV file. Afterwards, it is needed to specify in the `config.json` file that the new designed layout is used for the game

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(‘‘load_new_map’’: 1, 0 otherwise) together with the name of the new map in CSV configuration (‘‘map_file’’: ‘‘map_design.csv’’).

Time Constraints. In the configuration file config.json, ‘‘game_duration’’ controls the duration of playing time within an episode. The value is defined in seconds, so for a 5-minute mission the duration is set to 300.

Complexity. Minimap provides two levels of structural complexity: *simple* and *complex*. For example, in the configuration file above the environment of the game is complex (‘‘complexity’’: ‘‘complex’’). Basically, structural complexity is determined based on the number of obstacles and the arrangement of critical victims (see Fig. 4 and Section 3.1.1 for more details). However, it is worth noting that one can freely create new maps with different layouts and numbers of objects by following the steps described in the interface modification section.

Visibility. Minimap includes three levels of visibility to manipulate, namely the field of view (*fov*), map view (*map*) and full view (*full*). This degree of visibility is specified by ‘‘visibility’’ in the config.json file. With field of view, the individual can only observe part of the environment within their restricted vision area, while with map view, they can see the overview structure of the map in addition to their vision area (see Fig. 3 and Section 3.1.1 for more details). Conversely, the full view gives the player the whole view of the environment including where the victims are.

Time Victims Die. The moment victims die in the search and rescue task can be varied through the variable ‘‘victim_die’’. For example, in the above configuration file, critical victims die or disappear from the building after two minutes (120 seconds) of the game.

Delayed Time. The delay in the individual's actions, including victim saving and door opening, is integrated in Minimap by the variable ‘‘delayed_time’’, which is measured in milliseconds. In the example above, the delay period is set to one second, meaning that after the player executes an action of rescuing a victim, it takes one second for the action to take effect. The action occurs immediately with a delayed time of zero.

Number of Playing Episodes. The number of times an individual can perform the task is a variable in Minimap. For example, ‘‘max_episode’’ in the config.json file above controls the number of playing episodes ‘‘max_episode’’: 8 i.e., the default value is 8). This feature gives flexibility to experimenters to run experiments multiple times in order to examine the human learning process.

3. Experiment on Complexity and Opaqueness of Minimap

In the follow section, we demonstrate the capability of using Minimap to carry out behavioral experiments. The study varies in structural complexity and opaqueness under time constraints to investigate its effects on human decision-making behavior.

3.1. Methods

3.1.1. Experimental Design.

We conducted a between subjects study to investigate the effects of different levels of structural complexity and opaqueness on the decision behavior of individuals in the Minimap game. We specifically measured human performance in terms of different metrics concerning their behavior and decision making processes.

In the experiment, we considered two levels of opaqueness called Field of View (*FoV*) and Map view of the building (*Map*), and two levels of structural complexity, namely *Simple* and *Complex*. By following 2×2 factorial design we created four experimental settings that result from all possible combinations between the two levels of opaqueness and the two degrees of structural complexity.

The FoV setting provides participants with visibility of a 5×5 square viewing area around the participant's position, which is referred to the “field of view” or the “vision area”. Everything (i.e., doors, walls, obstacles, victims) outside that vision area is concealed (see Fig. 3a). In the Map view, by contrast, participants were aware of the doors and walls of the building in addition to the vision area as they were moving through the environment (Fig. 3b). In principle, human participants only receive incomplete information about the environment since neither the victim and blockage locations nor the total number of victims are disclosed to a rescuer.

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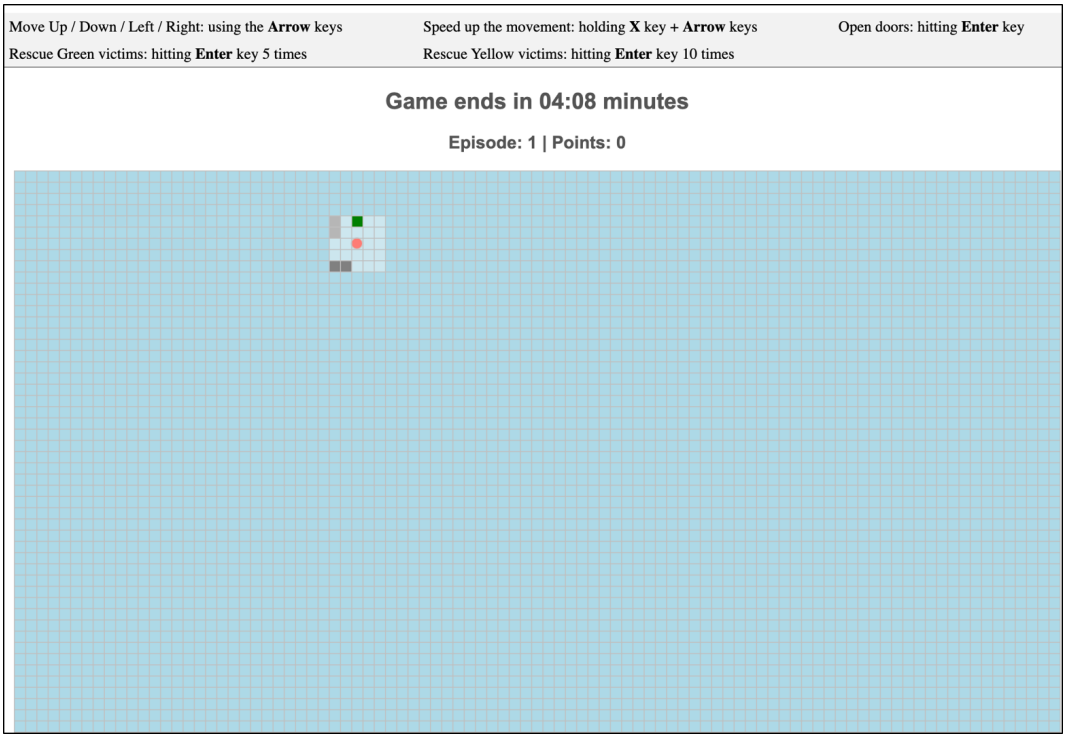

(a) Field of view (FoV)

(b) Map view

Figure 3: The illustration of Minimap with two different degrees of opaqueness: FoV and Map view.

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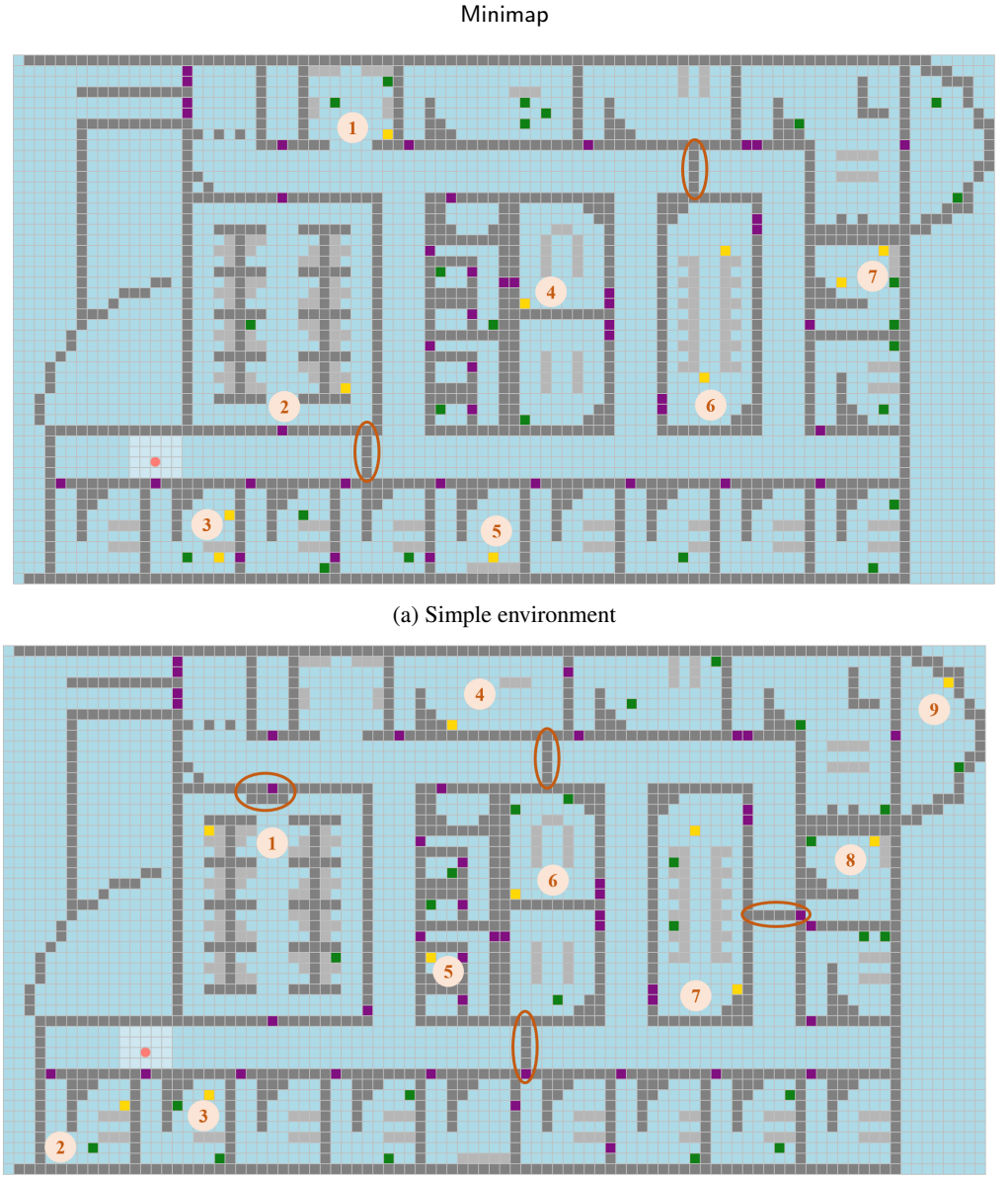

(a) Simple environment

(b) Complex Environment

Figure 4: Illustration of Different Levels of Structural Complexity

The distinction between levels of structural complexity is determined by the number of obstacles on the map and the placement of victims. In particular, additional blockages (encircled in red in Fig. 4) are placed in the middle of a path to restrict the individual from taking the blocked path, forcing them to look for longer paths to bypass the blockage. Consequently, the structural complexity of the task increases when there are more blockages. Furthermore, different levels of structural complexity differ in the arrangement of critical (yellow) victims. In the complex environment, the critical victims are scattered further away from the starting point, and hence they are more difficult to find. In particular, the yellow victims are spread out to nine rooms (Fig. 4b) in the complex environment while in the simple setting, they are located in seven rooms (Fig. 4a). Hence, it requires the rescuer's ability to search through the building. Overall, the design of the map could be *simple* (two blocks and victims located in seven rooms) or *complex* (four blocks and victims scattered in nine rooms) depending on the levels of complexity.

Regardless of the experimental conditions, the distribution over the number of regular (green) victims and critical (yellow) victims is the same across the conditions: 24 and 10 for regular and critical victims, respectively.

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Minimap

	Simple		Complex	
	#Participants	Mean Age ± Std.	#Participants	Mean Age ± Std.
FoV	48 (25 female)	(39.5 ± 10.03)	49 (22 female)	(41.8 ± 11.35)
Map	50 (22 female)	(37.9 ± 10.03)	48 (23 female)	(38.3 ± 10.91)

Table 1
Participant's distribution in each experiment condition

3.1.2. Participants

A total of 195 participants completed the study and were roughly evenly distributed under experimental conditions. All participants were recruited from Amazon Mechanical Turk for 30-45 minutes in a study pre-registered with the Open Science Framework[6]. The participants were compensated with a base payment of $1.50 and earned up to $3.00 in a bonus payment based on their accumulated score. The detailed descriptive statistics of the participants under the four conditions is reported in Table 1.

3.1.3. Procedure.

Each participant was randomly assigned to one of the four conditions of the mission. After giving their informed consent and receiving task instructions (see Appendix: Fig. A1, A2), the participants conducted the study consisting of two phases: (1) practice session and (2) main task. Subsequently, they completed a post-survey and demographic questionnaires.

In the practice session, all the participants were required to complete one-minute training to familiarize themselves with the controls. Then, they started a main task in which they were asked to complete eight five-minute Minimap episodes (that is, eight opportunities to perform the mission, and each mission episode lasts five minutes). The experimental conditions determined by visibility range and complexity of decision remained constant for each participant throughout the study. The participants started at the same location across the episodes.

3.1.4. Metrics and Measures.

We measured human performance with respect to the following metrics:

- **Effectiveness**: it is determined by the average proportion of rescued victims in each episode, which is correlated with the average points earned. The value ranges from 0 (least effective where no victims are rescued) to 1 (most effective where all victims are rescued).
- **Efficiency**: it is calculated by the ratio between earned points and the number of steps taken to rescue the last victims. The higher value of efficiency represents the individual's ability to find and save victims with shorter paths.
- **Coverage**: it is determined by the ratio of the number of uniquely visited locations to the total accessible locations on a given map. This metric captures the extent to which an individual explores the whole space.
- **Redundancy**: redundancy of the human trajectory is defined by the proportion of the number of revisited locations to the number of uniquely visited locations. The high value of redundancy means that the more frequent the individual makes mistakes in re-visiting locations that had been visited.

3.2. Results

Table 2 reports the average effectiveness, efficiency, coverage, and redundancy of human participants for each experimental condition of the task. As expected, we observe that, regardless of complexity, participants performed noticeably better in terms of effectiveness and efficiency when provided with more information about the environment (map view) compared to when restricted to a limited FoV. They were also able to explore more and made less

[6] https://osf.io/5gmsc/?view_only=b7b13bcae1da448e8c3a5d58ad976e34. According to the preregistration, we manipulated another level of structural complexity, which essential adds an additional obstacle (five obstacles), and victims were scattered in ten rooms instead of nine rooms as in the complex level. We did not observe any results standing out in this condition compared to the complex condition, and thus, we did not report them in this work.

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Minimap

	Opaqueness		Complexity	
	FoV	Map	Simple	Complex
Effectiveness	0.39 ± 0.22	0.56 ± 0.25	0.51 ± 0.24	0.45 ± 0.25
Efficiency	0.21 ± 0.09	0.30 ± 0.09	0.27 ± 0.1	0.23 ± 0.1
Coverage	0.41 ± 0.18	0.44 ± 0.17	0.44 ± 0.18	0.41 ± 0.18
Redundancy	0.33 ± 0.11	0.30 ± 0.11	0.31 ± 0.09	0.32 ± 0.13

Table 2
Descriptive statistics (mean ± standard deviation) on effectiveness, efficiency, coverage, and redundancy.

	Sum Sq	Mean Sq	NumDF	DenDF	F value	p	p.signif
episode	5.21	0.74	7	1337	101.70	$2.2e-16$	***
complexity	0.03	0.03	1	191	4.28	0.039	*
opaqueness	0.21	0.21	1	191	29.20	$1.9e-7$	***
episode:complexity	0.04	0.01	7	1337	0.76	0.619	
episode:opaqueness	0.26	0.04	7	1337	5.15	$8.4e-6$	***
complexity:opaqueness	0.00	0.00	1	191	0.23	0.629	
episode:complexity:opaqueness	0.05	0.01	7	1337	0.91	0.497	

$*$ denotes significant differences at $\alpha = .05$

Table 3
Results of the mixed ANOVA regarding the three-way interaction effect of opaqueness, complexity, and playing episode on humans' effectiveness.

redundant movements in with the higher level of environmental information. These observations hold true in light of the comparison between simple and complex settings, that is, human subjects achieved better performance in the simple environment than in the complex one.

However, it is clear that the average behavior provides a limited view of the variability and the dynamics of learning during the eight mission episodes. We therefore, further conducted a mixed ANOVA to analyze the two between-subjects factors (structural complexity and opaqueness) and the playing episode as within-subjects factor on humans' behavioral characteristics and their learning process with regard to each of the considered metrics.

3.2.1. Effectiveness

In Table 3, the ANOVA results revealed a non-significant three-way interaction effect between episode, complexity, and opaqueness $F(7, 1337) = 0.91, p = .497$, and a non-significant interaction effect between complexity and opaqueness $F(1, 191) = 0.23, p = .629$.

Opaqueness With regard to the simple main effect of opaqueness, the ANOVA results show that there was a statistically significant difference in the performance of humans at opaqueness levels $F(1, 191) = 29.20, p = 1.9e{-}7 < .05$. As shown in Fig. 5a, people were able to rescue more victims when provided with the map view over the FoV. Post hoc comparisons using Tukey's HSD corrections confirm that the mean proportion of victims rescued by participants playing in the map view setting ($M = 0.559, SD = 0.25$) was significantly higher than that of humans playing with the limited FoV setting ($M = 0.394, SD = 0.22$) at $p < .001$.

Fig. 5b demonstrates the learning curve of humans in terms of effectiveness across the eight episodes. In general, we see that no matter how much information was revealed, participants learned to improve by rescuing more victims as they performed the task more times. We also observe that the gap between the individuals' learning curve in Map view and those in FoV is widening over the eight playing episodes. Indeed, we found a significant interaction effect between opaqueness and episodes $F(7, 1337) = 5.15, p = 8.4e - 6 < .05$, suggesting that people with more information about the environment learned much faster and better compared to those who were limited to a vision area.

Structural Complexity Regarding the effect of structural complexity, the results of the ANOVA indicate that the effectiveness of the participants at the two levels of complexity was significantly different $F(1, 191) = 4.28, p = .039$.

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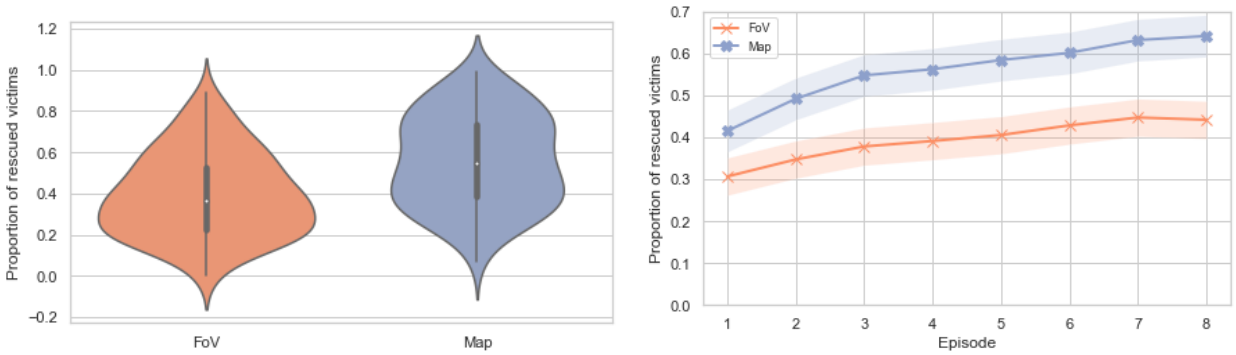
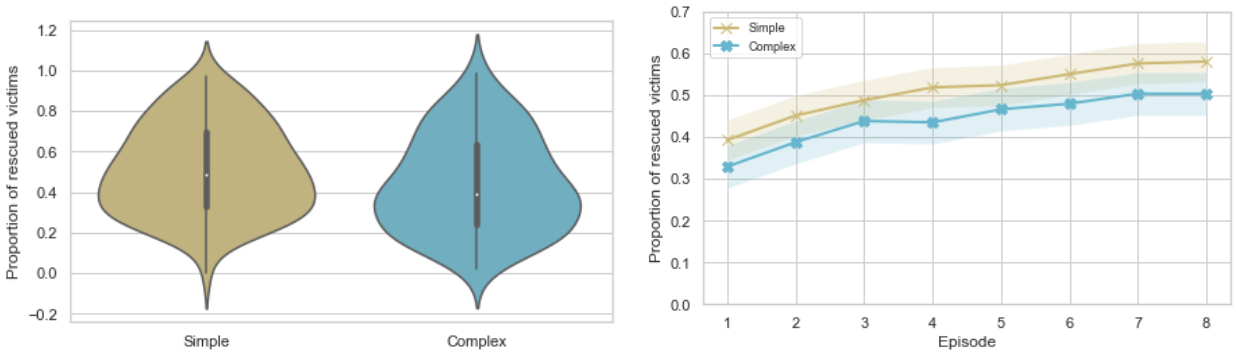


(a) Mean proportion of rescued victims at two levels of opaqueness.

(b) Mean proportion of rescued victims over time with 95% confidence interval error.

Figure 5: Effect of opaqueness on the humans' performance determined by the proportion of rescued victims.



(a) Mean proportion of rescued victims on two levels of complexity.

(b) Mean proportion of rescued victims over time with 95% confidence interval error.

Figure 6: Effect of structural complexity on the average human effectiveness measured by the proportion of rescued victims.

That is, Fig 6a show that people were more effective in rescuing victims in the simple environment than in the complex one. Post hoc analyzes with the Tukey HSD test confirm that the mean proportion of victims rescued in the simple environment ($M = 0.508, SD = 0.24$) was significantly higher than in the complex setting ($M = 0.445, SD = 0.25$) at $p = .039$.

From Fig. 6b we observe that participants' learning curve in either simple or complex environment slightly increases over time, suggesting that humans improve over time. However, we did not find a significant interaction effect between complexity and episode $F(7, 1337) = 0.76, p = .619$. The results is somehow counter-intuitive to the expected linear relation between environmental complexity and human learning performance over time. The results also suggest that the design of task complexity that relies on structural characteristic does not necessarily in a more complex task that can impair human decision learning.

3.2.2. Efficiency

Here we examined how the performance of humans in terms of efficiency is affected by the interaction effect among opaqueness, structural complexity and playing episodes. Table 4 reveals that there was no statistically significant three-way interaction between the effect of opaqueness, complexity and episode on humans' efficiency $F(7, 1337) = 0.39, p = .908$. We did not find evidence that the interaction effect between opaqueness and structural complexity was significant $F(1, 191) = 3.57, p = .061$.

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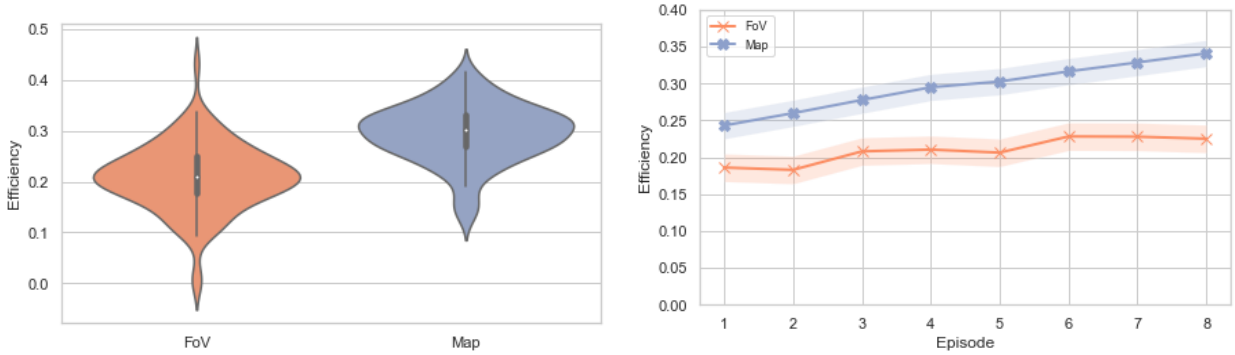
	Sum Sq	Mean Sq	NumDF	DenDF	F value	p	p.signif
episode	0.88	0.13	7	1337	27.55	$2.2e-16$	***
complexity	0.09	0.09	1	191	20.43	$1.1e-05$	***
opaqueness	0.48	0.48	1	191	104.66	$2.2e-16$	***
episode:complexity	0.06	0.01	7	1337	1.73	0.098	
episode:opaqueness	0.12	0.02	7	1337	3.77	0.0005	***
complexity:opaqueness	0.02	0.02	1	191	3.57	0.061	.
episode:complexity:opaqueness	0.01	0.00	7	1337	0.39	0.908	

$*$ denotes significant differences at $\alpha = .05$; . at $\alpha = .1$

Table 4
Results of mixed ANOVA on the effect of three-way interaction of opaqueness, complexity, and playing episode on human efficiency.



(a) Mean efficiency across two levels of opaqueness.

(b) Mean efficiency over time with 95% confidence interval error.

Figure 7: Effect of opaqueness on average human efficiency.

Opaqueness Fig. 7a clearly shows that participants performed the task more efficiently by saving more victims with fewer steps when provided with the map view compared to those presented with the FoV. We found a significant main effect of opaqueness on efficiency $F(1, 191) = 104.66, p = 2.2e-16$, and post hoc tests using Tukey's HSD also specified that the mean efficiency score in the map view ($M = 0.295, SD = 0.09$) was significantly higher than in the limited information FoV condition ($M = 0.21, SD = 0.09$) at $p < .0001$.

Furthermore, Fig. 7b shows that under the map view conditions, the efficiency of the participants progressively increased as more episodes passed, while that pattern does not occur for participants who were restricted to the FoV.

Structural Complexity Regarding the structural complexity, we found that there was a main effect of complexity upon the efficiency $F(1, 191) = 20.42, p = 1.1e-5$. Fig. 8a shows that participants performed the task more efficiently in the simple setting ($M = 0.271, SD = 0.1$) than in the complex environment ($M = 0.233, SD = 0.1$). Moreover, as would be expected, the effect of structural complexity on people's efficiency appears to be more pronounced than on the effectiveness, since the design of more complex environments requires participants to search for longer paths to work around the blockages.

Fig. 8b further shows the efficiency of participants over time. Interestingly, we noticed that the efficiency of individuals playing in the complex environment gradually increased throughout episodes while this trend was not apparent in the simple setting. That being said, there was no statistically significant difference in the efficiency score between the interaction effect of complexity and the playing episodes. The results once again suggest that increasing the structural complexity did not affect humans' learning ability in terms of efficiency over time.

3.2.3. Coverage

We measured the ability to explore the environment of humans in terms of coverage, that is determined by the proportion of covered areas to the entire space.

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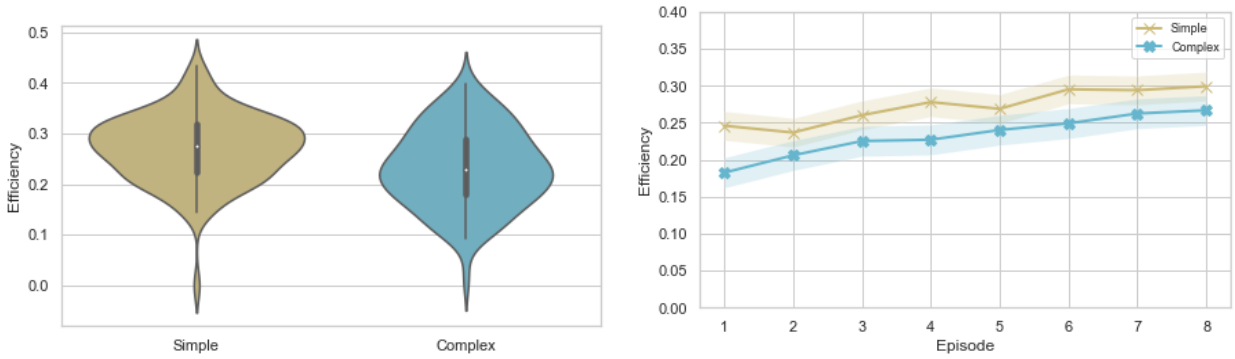

(a) Mean efficiency across two levels of structural complexity. (b) Mean efficiency over time with 95% confidence interval error.

Figure 8: Effect of structural complexity on the average humans' efficiency.

	Sum Sq	Mean Sq	NumDF	DenDF	F value	p	p.signif
episode	0.74	0.11	7	1337	31.81	0.000	***
complexity	0.00	0.00	1	191	0.64	0.423	
opaqueness	0.00	0.00	1	191	1.47	0.227	
episode:complexity	0.03	0.00	7	1337	1.26	0.265	
episode:opaqueness	0.02	0.00	7	1337	0.77	0.613	
complexity:opaqueness	0.00	0.00	1	191	0.03	0.870	
episode:complexity:opaqueness	0.01	0.00	7	1337	0.44	0.878	

$*$ denotes significant differences at $\alpha = .05$

Table 5
Mixed ANOVA results for the effect of the three-way interaction of opaqueness, complexity, and playing episodes on human coverage.

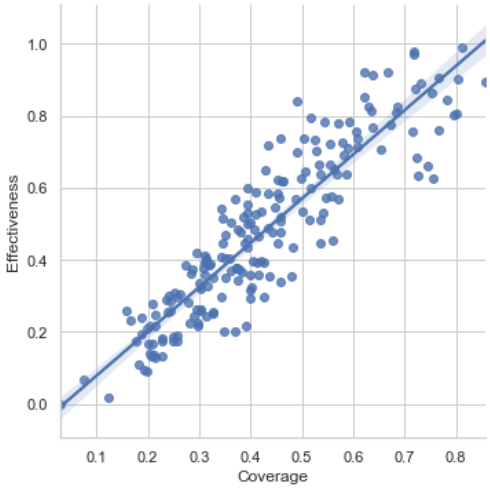

Figure 9: Scatter plot showing the relationship between coverage and effectiveness.

The results of the ANOVA in Table 5 reveal that there was no statistically significant interaction between the effects of opaqueness, complexity, and episode on coverage $F(7, 1337) = 0.44, p = .878$. Likewise, we did not find significant main effects of opaqueness $F(1, 191) = 1.47, p = .227$ nor structural complexity ($F(1, 191) = 0.64, p = .423$) on the

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	Sum Sq	Mean Sq	NumDF	DenDF	F value	p	p.signif
episode	0.09	0.01	7	1337	2.09	0.042	*
complexity	0.00	0.00	1	191	0.10	0.748	
opaqueness	0.03	0.03	1	191	5.37	0.022	*
episode:complexity	0.10	0.01	7	1337	2.22	0.031	*
episode:opaqueness	0.08	0.01	7	1337	1.76	0.092	.
complexity:opaqueness	0.00	0.00	1	191	0.13	0.717	
episode:complexity:opaqueness	0.02	0.00	7	1337	0.52	0.821	

$*$ denotes significant differences at $\alpha = .05$. at $\alpha = .1$

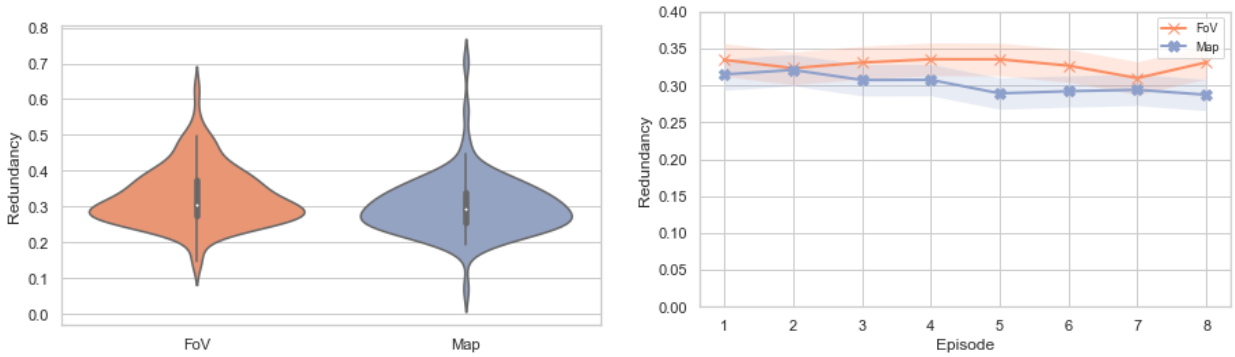
Table 6
Results of mixed ANOVA on the effect of the three-way interaction of opaqueness, complexity, and playing episodes on human redundancy.

coverage. The results suggest that human ability to explore the environment is not influenced even when there is an increase in opaqueness and structural complexity. However, we found that individual coverage was significantly affected by the play of episodes $F(7, 1337) = 31.81, p = .000$, indicating that as the individual played more episodes, the proportion of areas they explored increased accordingly.

In addition, we analyze the association between this coverage metric and human performance in terms of effectiveness. The scatter plot in Fig. 9 shows that there is a strong, positive association between the participants' coverage rate and their effectiveness determined by the proportion of rescued victims (Spearman rank correlation: $r_s = 0.91, p = 1.81e{-}75$). This clearly suggests that individuals with higher coverage rate are more likely to save more victims, and likewise those with lower coverage rate tend to be less effective.

3.2.4. Redundancy

We analyze the redundancy of the movements made by the participants while navigating through the environment at different levels of opaqueness and structural complexity throughout the playing episodes. From the mixed ANOVA in Table 6, there was no significant three-way interaction between the effect of opaqueness, complexity, and episode on the redundancy rate $F(7, 1337) = 0.52, p = .821$. We did find the significant main effect of episode $F(7, 1337) = 2.09, p = .042$, opaqueness $F(1, 191) = 5.37, p = .022$, and the interaction effect between structural complexity and episodes $F(7, 1337) = 2.22, p = .03$ on human redundancy.



(a) Mean redundancy across two levels of opaqueness.

(b) Mean redundancy over time with 95% confidence interval.

Figure 10: Effect of opaqueness on average human redundancy.

Opaqueness Fig. 10a shows the proportion of redundant visits made by participants across the two levels of opaqueness. We observe that participants presented with the map view made fewer redundant movements than those in the FoV. The statistical tests further support this observation by indicating that there was a significant effect of opaqueness on the redundancy rates $F(1, 191) = 5.37, p = .022$. Post-hoc comparisons using Tukey's HSD specified

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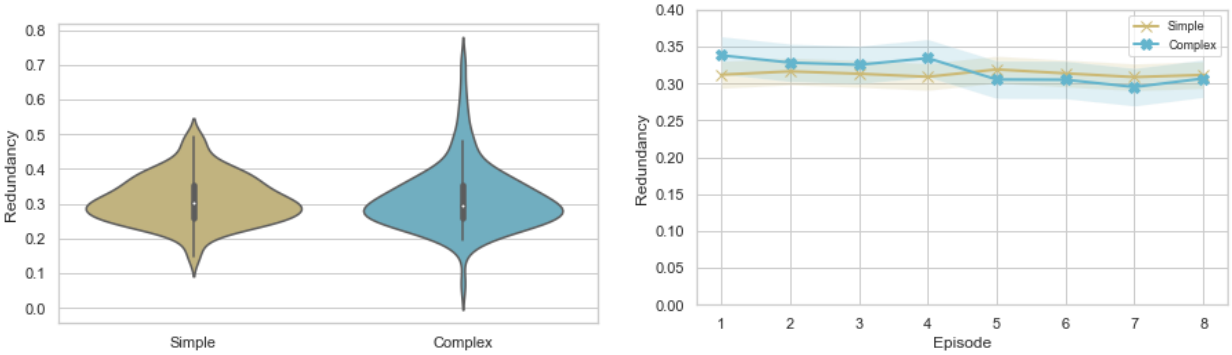
that the mean redundancy rate made by participants with restricted information in FoV ($M = 0.329, SD = 0.11$) was significantly higher than those provided with the map information ($M = 0.302, SD = 0.11$).



(a) Mean redundancy in two levels of structural complexity.

(b) Mean redundancy over time with 95% confidence interval.

Figure 11: Effect of structural complexity on average human redundancy.

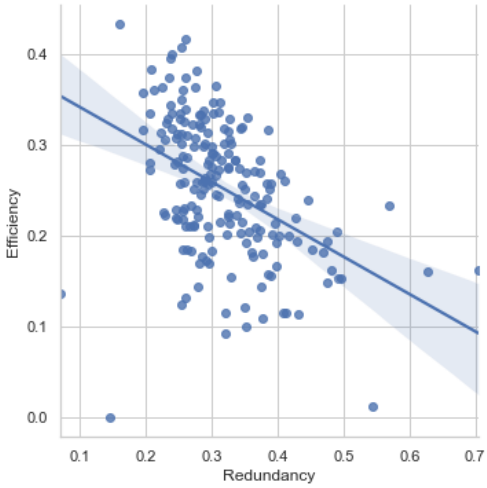


Figure 12: Scatter plot showing the relationship between redundancy and efficiency.

Structural Complexity Furthermore, Fig. 10b shows that the redundancy rate was slightly reduced during the first five episodes under the map view condition, while in the FoV setting the redundancy appeared to be unchanged across the episodes. We however, did not find the statistical difference in the redundancy rate between the two levels of opaqueness throughout playing episode. In general, the results suggest that people tend to make better decisions by taking fewer redundant visits when given more about the environment, compared to those who are not. We also learn that despite the availability of disclosed information, people were unable to learn to reduce their redundant movements over time as the more episodes they experienced.

Regarding the structural complexity, we did not find the significant difference between the participants' redundancy rate in the simple and complex environments, but there was a statistically significant interaction between the effect of structural complexity and episodes on the proportion of redundant visits made by participants $F(7, 1337) = 2.22, p = .03$. In particular, post-hoc Tukey HSD comparisons specify that the mean rate of making redundant visits in the complex environment at the first episode ($M = 0.338, SD = 0.15$) is significantly higher than that at the seventh episode ($M = 0.295, SD = 0.08$) with $p = .016$. The redundant rate of participants playing in the simple environment appears to be unchanged.

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Perhaps not surprisingly, we found that there is a medium negative association between the rate of redundant visits made by participants and their performance in terms of efficiency (Fig. 12). Concretely, the Spearman rank correlation resulted in $r_s = -0.49$, $p = 3.69e{-}13$. As such, the more redundant revisits made while navigating the environments, the less efficient one becomes in rescuing the victims.

4. Discussion and Conclusions

In this paper, we introduced Minimap, an interactive browser-based microworld that is built to help researchers study many aspects of human behavior in dynamic decision-making (DDM) problems and improve our understanding of how people make decisions in such environments. The main motivation behind the development of Minimap was to empower behavioral researchers with a tool to systematically manipulate or vary the different characteristics of a DDM task, so they can address specific research questions in various research contexts. That is, Minimap can be used to simulate many real-world dynamic situations without losing their vital characteristics. As described in 2, Minimap itself integrates all four essential characteristics of a DDM task: *dynamics*, *complexity*, *opaqueness*, and *dynamic complexity*. Moreover, such characteristics can be easily manipulated within Minimap. For example, one can easily vary the level of *opaqueness* using different visibility configurations offered in Minimap; or the length of feedback delays (*dynamic complexity*) is also adjustable by altering the time periods that an action takes effect (e.g., an action of rescuing a victim can take effect immediately or after a certain time intervals).

Furthermore, Minimap is designed to provide experimenters with control over creating various scenarios (cover story) and different structures of a task. For instance, Minimap, by default, is contextualized in a urban search and rescue mission, but in general it can be turned into any mission scenarios such as treasure hunting, wildlife conservation, fire forest or sea rescue. The graphical design of the task is highly customizable without the need for specialized technical knowledge. Modifications can range from minor to major changes, depending on the analogies between a new scenario and the search and rescue mission. However, the graphical appearance and user control are fully modifiable for DDM research needs.

Crucially, unlike most existing microworlds, Minimap is built to make use of the full capabilities of modern browsers for interactivity and hence offers great flexibility to experimenters in the definition of experimental setups (i.e., study human interactions at different time intervals) as well as online data collection with diverse subject populations. The Minimap source code is open and can be downloaded for free at `https://github.com/DDM-Lab/MinimapInteractiveDDMGame`.

We have also demonstrated a concrete research example of using Minimap to examine the effects of structural complexity coupled with opaqueness of the system on human decision making performance. The study is contextualized in a simulated search and rescue mission, motivated by the fact that people in such a disaster environment often have to deal with a complex topological structure of the setting, accompanied by uncertain conditions when performing rescue operations. Our findings show that: (1) humans performed more effectively and efficiently when they were provided with the Map view than with FoV; (2) humans made significantly less redundant movements in the Map view compared to FoV; (3) increased structural complexity did not affect humans' learning in terms of effectiveness and efficiency, but did impact humans' mistakes in making redundant visits over time; and (4) human coverage, that is, the ability to explore the environment, was not affected even when opaqueness and structural complexity increase. These findings suggest that when people have more information about the environment, they perform the task more effectively and efficiently than those who do not. The results further provide evidence that decreasing the opaqueness of situations helps humans reduce their redundant moves to the place they had been. Additionally, the experiment specifies that increased structural complexity did not affect human learning performance over time, which contradicts the intuitive expected linear relation between the complexity level and the learning process. Finally, neither structural complexity nor opaqueness influence individual's the ability to explore the environment that is captured by the coverage metric.

Implications

We believe that the experimental results can inform the design of the complexity of tasks in decisions under uncertainty. Concretely, we have learned from this work that complex structural characteristics alone may be inadequate to determine the complexity of a task in practice and that other elements, such as high interconnectivity, changing dynamically over time, or the difficulty of identifying the cost and benefits of each alternative (ie, complexity of decision (Nguyen and Gonzalez, 2020)) can play a role in dictating the complexity of the task.

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Furthermore, the objective of building AI systems should not ultimately be to solve specific tasks autonomously, but rather to focus on the development of interactive assistant systems where humans and machines work together taking advantage of their respective strengths (Forbus and Hinrichs, 2006; McNeese, Demir, Cooke and Myers, 2018). Hence, it is crucial to first understand human behavior as insights obtained from human subjects play integral role in informing and providing suggestions for AI system designs (Lake, Ullman, Tenenbaum and Gershman, 2017; Botvinick, Ritter, Wang, Kurth-Nelson, Blundell and Hassabis, 2019). As such, we are convinced that our findings provide useful implications for training and building advisory support systems that attempt boost human performance in many fields in which they must cope with structural complexity associated with environmental opaqueness. For example, our results show that the extent to which an individual explores the space and makes a mistake in taking redundant movements predicts the performance of the task. Hence, the system can leverage such in-game process observations to offer a timely intervention by giving humans advice based on the observed process metrics. As a follow-up research, this observation may also point to an opportunity to explore how AI agents can be adapted to humans as teammates in human-machine teaming tasks.

Finally, the use of Minimap is aimed to transcend the scope of individual tasks, provided that microworlds in DDM research literature tend to present a decision-making problem of individuals rather than teams (Brehmer and Dörner, 1993; Gonzalez et al., 2005b). Minimap is under active development to feature a team environment wherein multiple players are assigned to different roles to play the game together at the same time. Moreover, considering that a number of off-the-shelf machine learning libraries and agent-based models (Nguyen, Phan and Gonzalez, 2022) are written in Python, Minimap's architecture relying on Python's API framework makes it even easier to incorporate any of these libraries in an attempt to connect and build a synthetic teammate. With that, the use of Minimap is expected to increase in the future given rapid advancement of research in the context of human-agent teaming.

Transparency and Openness

Minimap is provided as a free and open-source application, which can be accessed at `https://github.com/DDM-Lab/MinimapInteractiveDDMGame`. All human data and all scripts used for the analyses presented in this manuscript are also available on Github: `https://github.com/DDM-Lab/HumanExperimentMinimap`.

Acknowledgements

This research was sponsored by the Defense Advanced Research Projects Agency and the Army Research Office and was accomplished under Grant Number W911NF-20-1-0006.

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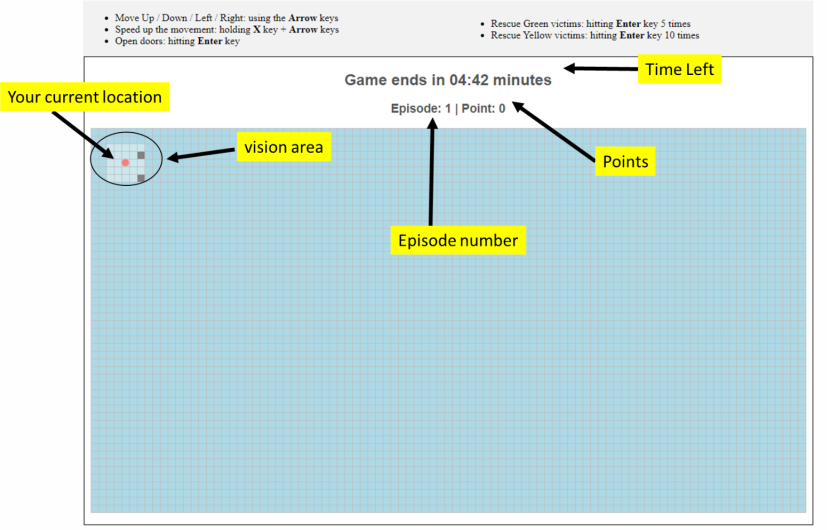
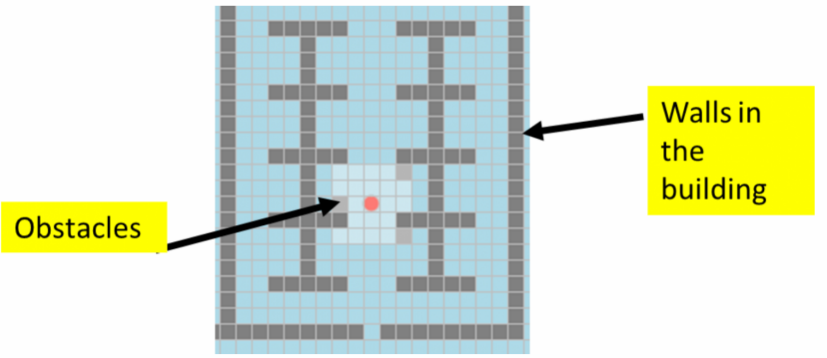
Appendices

"Minimap" rescue mission: Structure of the game and Goals

In this task your goal is to rescue as many victims as possible from inside a building within a time limit. The Figure shows the interface of this game.



Your location within the building is identified as a pink dot on the screen. The building has many rooms, doors, walls and obstacles around. Sometimes, you will only be able to see the walls and obstacles around you as shown in this section of the building:



but other times the walls and the obstacles will be concealed, and you will only be able to see an area of 5x5 squares around you ("vision area"). Thus, you need to search around to find the victims to rescue, which are identified as Green and Yellow squares.

You will perform a training scenario (i.e., Episode: 0), and 8 real episodes where you can earn points. The training episode lasts 1 minute, and each of the real episodes (Episode 1… 8) last 5 minutes. The time left in the episode will be shown in the screen (i.e., Time Left). You will earn points by rescuing the victims and these points accumulate (i.e., "Points") during each of the real episodes. Your bonus will be calculated in the following way. The total points earned across all 8 episodes will be divided by 8, and we will pay 1 cent for each transformed point. For example, if you earned 2800 points during the 8 episodes, that divided by 8 = 350, and 1 cent per adjusted point gives us $3.50.

Figure A1: Instructions on the structure and objective of the game

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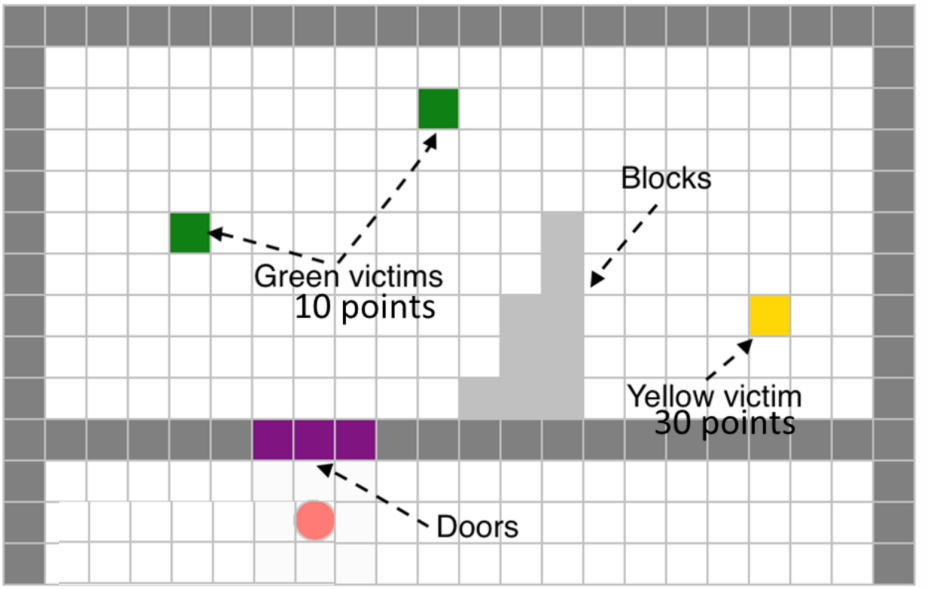
"Minimap" rescue mission: Structure of the Game and Goals

The figure below shows some of the elements of the game. In the building there are two types of victims: The Green victims are less injured, and they will give you 10 points, if rescued. The Yellow victims are more injured, and they will give you 30 points. Yellow victims die at minute 4: they will disappear from the building, and Green victims will not die during the mission.



You will find blocks around the building, and you will need to walk around those blocks and walls to be able to rescue the victims. Also, rooms have closed doors which you will need to open in order to enter. To rescue victims or to open doors you need to be in a square right next to the victim or the door. Here are the keyboard controls for the game:

1) Move Up / Down / Left / Right using the Arrow keys.
2) Speed up the movement: Holding X key + Arrow keys
3) Open doors: hitting Enter key
4) Rescue Green victims: hitting Enter key 5 times
5) Rescue yellow victims: hitting Enter key 10 times.

To help you understand the task, you will do the practice round: Episode 0 on the next screen. You will begin a real episode (Episode 1), after the practice round.

After clicking the Next button below, you will start the practice session.

IMPORTANT: Please do NOT refresh the page or attempt to navigate backward at any point in this study. Doing so will void your response and prevent us from issuing your payment.

Figure A2: Instructions for using the controls to play the game.

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